



APPROVED

By John Ozoga III at 11:29 am, Nov 07, 2023

October 27, 2023

Michigan Department of Environment, Great Lakes and Energy
Materials Management Division
Cadillac District Office
120 W. Chapin Street
Cadillac, Michigan 49601
Submitted via email

Attn: John Ozoga, III, Assistant District Supervisor
ozogaj@michigan.gov

Subject: **Gas Migration Monitoring Plan**
Wexford County Landfill, LLC
Manton, Michigan
WDS 470336

Dear Mr. Ozoga:

Attached please find the above referenced Gas Migration Monitoring Plan (GMMP) for the Wexford County Landfill, LLC. The GMMP was prepared by Civil & Environmental Consultants, Inc. This submittal includes the following information:

- GMMP Review Checklist, as requested by Michigan Department of Environment, Great Lakes, and Energy (in Excel format)
- GMMP (in PDF format)
- Summary table of the past five years of perimeter landfill gas monitoring data (in PDF format)

Should you have any questions regarding the attached report, please contact me at (231) 597-8553.

Sincerely,

A handwritten signature in black ink, appearing to read 'Chris Gee', written over a horizontal line.

Chris Gee
General Manager

att: GMMP Review Checklist, electronic format
Gas Migration Monitoring Plan, electronic and hard copy format
Historical Perimeter Landfill Gas Monitoring Data, electronic format

cc: Wexford County Landfill, LLC Operating Record (hard copy of the GMMP)
J. Collins, jcollins@gflenv.com
T. Craig, tamicraig@gflenv.com
C. Powrozek, cpowrozek@cecinc.com
S. Paulsen, spaulsen@cecinc.com

October 27, 2023

Chris Gee (cgee@gflenv.com)
Wexford County Landfill, LLC
990 North Mackinaw Trail (US-131)
Manton, Michigan 49663
Submitted via e-mail

Dear Mr. Gee:

Subject: Gas Migration Monitoring Plan
Wexford County Landfill, WDS # 470336
CEC Project 331-878

Civil & Environmental Consultants, Inc. (CEC) has prepared an updated Gas Migration Monitoring Plan (GMMP) for the Wexford County Landfill (WCL) in Manton, Michigan. The GMMP was developed for explosive landfill gases at the WCL in accordance with Part 115 of the Michigan Natural Resources and Environmental Protection Act, 1994 PA 451, as amended, and the administrative rules promulgated pursuant to the Act.

This submittal includes the following information:

- GMMP Review Checklist
- GMMP
- Past five years of perimeter landfill gas monitoring data
 - Three exceedances were noted in perimeter gas probes in the past 5 years.
 - On October 29, 2021, GP-4R reported methane above the lower explosive limit (LEL). Three barholes were punched to three feet below ground surface to the east of the gas probe, between the gas probe and the property boundary. No detections were reported in the three barholes.
 - On October 12, 2022, two perimeter gas probes (GP-4R and GP-5R) had exceedances noted based on a review of the quarterly perimeter landfill gas monitoring data for the past five years. At the time of the perimeter landfill gas monitoring, the gas extraction and flare system were turned off for repairs. The system was back on the following week, and verification monitoring was conducted on October 19 and October 20, 2022, and the field staff recalls no exceedances. Unfortunately, the documentation of the field readings from the October 19 and October 20, 2022, verification event

has not been located, but subsequent monitoring events have reported no detections of methane in GP-4R or GP-5R, indicating that the detections reported on October 12, 2022 were directly related to the inactivity of the gas extraction and flare system.

- o The type of electronic gas detector in each facility structure is a Mine Safety Appliances (MSA) Ultima Gas Monitor.
- o Ambient air monitoring of the facility structures has been conducted on a quarterly basis for at least the past 5 years. No exceedances have been reported at or above twenty-five percent of the LEL.

This *Gas Migration Monitoring Plan, Wexford County Landfill*, has been prepared by a qualified groundwater scientist with Civil and Environmental Consultants, Inc. of Toledo, Ohio. If you have any questions, please contact the undersigned at (419) 724-5281 or Mr. Chris Gee at (231) 258-9030.

Sincerely,

CIVIL & ENVIRONMENTAL CONSULTANTS, INC.



Carolyn E. Powrozek, C.P.G.
Project Manager, Senior Geologist



Sean C. Paulsen, P.G.
Principal

Enclosures: Gas Migration Monitoring Plan Review Checklist (Excel format via e-mail only)
Gas Migration Monitoring Plan (PDF format via e-mail and hard copy for Operating Record)
Historical Perimeter Landfill Gas Monitoring Data (PDF format via e-mail only)

cc: Wexford County Landfill Operating Record (hard copy)
Tami Craig, GFL Environmental tamicraig@gflenv.com
John Collins, GFL Environmental, jcollins@gflenv.com

GAS MIGRATION MONITORING PLAN

**WEXFORD COUNTY LANDFILL
WDS NO. 470336**

Prepared for:

**WEXFORD COUNTY LANDFILL, LLC.
MANTON, MICHIGAN**

Prepared by:

**CIVIL & ENVIRONMENTAL CONSULTANTS, INC.
TOLEDO, OHIO**

CEC Project 331-878

OCTOBER 2023



Civil & Environmental Consultants, Inc.

TABLE OF CONTENTS

1.0	INTRODUCTION.....	1
1.1	Site Location and Background Information.....	1
1.2	Landfill Gas Control System	2
2.0	REGULATORY REQUIREMENTS.....	2
2.1	Allowable Gas Levels	2
2.2	Contingency for Exceeding Allowable Levels	2
3.0	GAS MIGRATION MONITORING	3
3.1	Facility Building Structures	3
3.2	Facility Perimeter Monitoring.....	3
3.2.1	Gas Probe Depth and Design	3
3.2.2	Ambient Air and Barhole Monitoring.....	4
4.0	SAMPLING EQUIPMENT AND METHODS	4
4.1	Facility Structure Monitoring Procedures.....	4
4.2	Permanent Monitoring Probe Sampling Procedures.....	5
4.3	Barhole Sampling Procedures.....	5
5.0	RECORDKEEPING.....	5
5.1	Quarterly Reporting	5
5.2	Plan Revisions.....	6

FIGURES

Figure 1 – Sample Location Map

TABLES

Table 1 – Summary of Gas Migration Monitoring

APPENDICES

Appendix A – Perimeter Gas Probe Logs

Appendix B – Example Gas Migration Monitoring Field Form

1.0 INTRODUCTION

The Wexford County Landfill (WCL) is a licensed Type II solid waste landfill that is owned and operated by Wexford County Landfill, LLC. The site is in a rural setting, located at 990 North Mackinaw Trail, Manton, Michigan. This Gas Migration Monitoring Plan (GMMP) was developed for explosive landfill gases at the WCL in accordance with Part 115 of the Michigan Natural Resources and Environmental Protection Act, 1994 PA 451, as amended, and the administrative rules promulgated pursuant to the Act. This GMMP provides background information, monitoring methods, frequency, and recordkeeping requirements for explosive gas migration monitoring at the landfill.

1.1 Site Location and Background Information

The WCL is in the West ½ of the Northwest ¼ of Section 34 and East ½ of Section 33, Township 23 North, Range 9 East, of Cedar Creek Township, Wexford County, Michigan. Access to the site is from North Mackinaw Trail, which is also identified as United States Highway 131 (US-131). The facility consists of an existing Type II Landfill, consisting of 196.4 acres, that has two separate units (East Unit and West Unit) with active portions of the East Unit Cells B through G, Cell G3, Cell H-East, and Cell J, closed portions of the East Unit Cells B, C, D/E, F, G-Phase 1, Cell G-Phase 2, and J, and future unconstructed of East Unit Cell H-West and Cell I as well as West Unit Cells 1 through 9. The landfill layout, existing perimeter gas probes and site features are shown on **Figure 1, Sample Location Plan**.

The site is set among rural wooded areas and the land in Cedar Creek Township. Properties surrounding WCL are currently zoned as rural residential. The WCL property is zoned as Industrial. Other than the landfilled area, the site topography is characterized as moderately rugged terrain. The “knob and swale” topography of the area is typical of terminal moraines. From the surface downward, the four main geologic units, including:

- Upper Sand and Gravel Unit: sand and gravel with varying amounts of clays, silts and gravels. Where saturated, this unit is known as the uppermost aquifer.
- Clay Aquitard: sandy clay to lean clay with trace amounts of sand and gravel. This unit varies in thickness from three to greater than thirty feet.
- Lower Sand and Gravel Unit: sand and gravel with silty sand and gravel beds in the upper portion of the unit. This is the deepest unit encountered during site hydrogeologic investigations.
- Bedrock: Bedrock deposits of sandstone, shale, and limestone of the Saginaw Formation.

The WCL is constructed over the Upper Sand and Gravel Unit, which is the uppermost aquifer. The piezometric surface occurs near the base of the Upper Sand and Gravel Unit, between 1,223 feet (ft) and 1,232 ft above mean sea level (amsl). The uppermost aquifer is recharged via infiltration from ground surface. The landfill floor ranges from 1,330 ft to 1,400 ft amsl. There is

at least 100 ft of vertical isolation between the bottom of the landfill and groundwater. These geologic conditions significantly limit the potential for horizontal methane migration at the facility.

1.2 Landfill Gas Control System

The WCL gas collection and control system (GCCS) is operated in accordance with the Site's GCCS plan, which provides detailed descriptions regarding the compliance with air pollution regulations. In general, WCL uses gas extraction wells to remove landfill gas generated within the landfill. Currently there are 41 gas extraction wells present and 5 new gas extraction wells anticipated before the end of 2024. Additional gas extraction wells and passive vents will be added as appropriate. Currently, landfill gas is directed to the utility flare on Site as needed.

2.0 REGULATORY REQUIREMENTS

Rule 433(2) and (3) require that the owner and operator implement a routine methane monitoring program to verify that the allowable gas levels are not exceeded. The type and frequency of monitoring is required to be based on soil conditions, hydrogeological and hydraulic conditions surrounding the facility, the location of facility structures, and distance to property boundaries. The minimum frequency for quarterly monitoring is required by Rule 433(3).

2.1 Allowable Gas Levels

Rule 433 addresses the need for monitoring methane gas concentrations in facility structures and at the property perimeter. The concentration of methane is required to be less than 25 percent of the lower explosive limit (LEL) for methane (1.25 percent methane) in facility structures, excluding gas control or recovery system components and the leachate collection system. At the property perimeter, the methane concentration is required to be less than the LEL (5.0 percent methane).

2.2 Contingency for Exceeding Allowable Levels

In the event that methane gas levels exceed the limits specified in Section 2.1, the owner and operator shall do all the following:

- Immediately take all necessary steps to verify protection of human health.
- Notify the site management, Regional Landfill Gas Program Manager, and Regional Environmental Compliance Manager.
- Notify the Michigan Department of Environment, Great Lakes and Energy (EGLE) at the Material Management Division (MMD) Cadillac district office.
- Within seven days of exceedance, place in the operating record the methane gas levels detected, and a description of the steps taken to protect human health.
- Within 60 days of exceedance, implement a remediation plan for the methane gas releases, place a copy of the plan in the operating record, and notify the EGLE MMD that the plan

has been implemented. The plan shall describe the nature and extent of the problem and the proposed remedy.

3.0 GAS MIGRATION MONITORING

Landfill gas migration reporting will include monitoring of facility structures and at the property boundary on a quarterly basis. This section describes monitoring requirements for the WCL.

3.1 Facility Building Structures

Landfill gas monitoring will include continuous monitoring in three facility buildings and any additional buildings constructed on a quarterly basis. Permanent gas detectors, such as Mine Safety Appliances (MSA) Ultima Gas Detectors or equivalent, are currently located in the scalehouse-office, maintenance building, and leachate management building. The Site is coordinating the installation of a permanent gas detector at the deep well building. Currently there are no offsite structures that have permanent gas detectors. Any new buildings constructed at the landfill facility will also be equipped with permanent gas detectors following construction. The location of the existing buildings and building gas monitors relative to the waste unit is shown in Figure 1. The gas detectors provide continuous gas monitoring and are built to sound an alarm before 25 percent (typically in the 15-20 percent depending on manufacturer's specifications) of the LEL is detected. Detectors will be installed and tested in accordance with manufacturer's recommendations. At a minimum, detectors will be tested on a quarterly basis. Detectors that are not functioning properly or are beyond their service date (typically 10 years from the date that they are placed into service) will be replaced by the same model or equivalent.

In addition to continuous alarm monitoring, quarterly monitoring of the LEL in ambient air within facility building will also be conducted using a portable gas meter, as described by Section 4.0. If conditions warrant, a more frequent schedule will be proposed.

3.2 Facility Perimeter Monitoring

The current perimeter monitoring plan consists of monitoring of seven existing gas probes, seven future gas probes (to be installed prior to construction of the West Unit), and four barhole locations along the existing site perimeter. The gas probes are installed around the perimeter of the landfill with a minimum of one on each side. A summary of the existing monitoring probes is included in **Table 1, Gas Migration Monitoring Summary** and the locations are shown on Figure 1.

3.2.1 Gas Probe Depth and Design

Monitoring probes will be placed in the Upper Sand and Gravel Unit and extend to an approximate depth of ten feet below the waste grade of cell nearest the gas probe as required in The Act (Part 115 of NREPA), Section 11512b (3). All seven existing probes extend at least ten feet below base grades of currently constructed cells; however, one gas probe (GP-8), indicates that it will not be a minimum of ten feet below waste grade for future cell waste grades. In general, the probe screens

are constructed of slotted schedule 40 PVC and the screened portion extends from approximately five feet below ground surface to the total depth of the probe. The probes depths were selected to match the maximum depth of waste adjacent to each probe.

Probes are be capped, and a port is installed to facilitate attachment of the portable methane gas detector during monitoring. Number 30 coarse sand or 10/20 sand pack (filter sand packs) was used to backfill the annulus surrounding the probe screen to a depth of six inches above the top of screen. The backfill above the filter sand packs consists of a bentonite seal to approximately 3 feet below ground surface. A six-inch square steel protective casing, equipped with a hinged locking cover, extends to approximately three feet above ground surface and is set within a concrete pad and cap to stabilize the protective casing and to promote drainage of surface water away from the base of the probe.

The existing gas probe logs are provided in **Appendix A, Perimeter Gas Probe Logs**.

3.2.2 Ambient Air and Barhole Monitoring

Four temporary barhole locations are monitored at WCL due to distance between the current cell limit and the nearest property boundary. Barhole locations BH-1 and BH-4 are located to the north of Cell I, within the solid waste boundary of Cell I. Barhole locations BH-2 and BH-3 are located within the solid waste boundary of future Cells 2 and 3. Monitoring will be conducted at these four temporary locations until perimeter probes 9 through 15 are installed. Perimeter probes GP-9 through GP-15 will be installed prior to construction of any cells in the western expansion area.

During each monitoring event, a barhole (or temporary hole) will be created to a depth of 2 ½ to 3 feet (depending on the estimated depth of saturated sediments) below the ground surface using a steel plunger bar. After punching the plunger barhole, the plunger bar will be removed. Following removal, an appropriate length of polypropylene tubing that is either open or perforated on the bottom will be inserted into the temporary barhole and the barhole will be sealed at ground surface with low-permeability materials (bentonite or nearby surface soils). The portable gas detector will be attached to the end of the polypropylene tube and readings will be collected with the portable gas detector. The procedure for gathering readings from the temporary probe will be similar to the procedure for monitoring a permanent monitoring probe (as described in Section 4.2 below). Once a stabilized reading is observed on the detector, it will be recorded.

4.0 SAMPLING EQUIPMENT AND METHODS

4.1 Facility Structure Monitoring Procedures

The facility structures are equipped with continuous gas detectors, (MSA Ultima Gas Detectors), that are located within the breathing zone. In addition to the continuous gas monitors, the facility structures will be monitored on a quarterly basis using a portable gas meter, such as a Landtec GEM, RKI GX-3R, or equivalent. The portable gas meter will be calibrated to read percent methane or percent LEL and will be capable of pumping a representative sample from the probe.

The portable gas meter will be calibrated in accordance with the manufacturer's recommendations. Because methane gas is lighter than air and thus rises, readings will be taken along the ceiling near the wall in closest proximity to the landfill. The portable gas meter must be allowed to pump and analyze a representative sample sufficient to guarantee valid results. The manufacturer's instructions should be used to determine the correct sampling duration.

Following each quarterly monitoring event, the type of permanent gas detector installed at each facility structure, as well as the make and model of portable gas meter, will be included with the calibration records and quarterly field data sheets, and will be placed in the site's operating record.

4.2 Permanent Monitoring Probe Sampling Procedures

The monitoring probes will be tested using a portable gas detector. Immediately after approaching the probe and connecting to the sampling port, an air sample from the probe will be withdrawn per manufacturer's specifications for that meter, allowing the reading to stabilize for a minimum of thirty seconds before recording the measurement. The percentage of methane (or percentage of the LEL) will be recorded along with other pertinent data (e.g., weather, oxygen concentrations). Should an initial reading indicate an exceedance of the LEL, the probe will be re-tested within 30 days and both the initial and re-test values will be reported. Once the methane detector has stabilized, the reading will be recorded on the field data sheet.

4.3 Barhole Sampling Procedures

If conditions warrant additional sampling (e.g., elevated methane levels in the probe or distressed vegetation at the landfill property boundary indicating gas migration), a temporary barhole can be created to monitor gas readings in the near surface soils. The barhole should be punched to a depth of 2 ½ to 3 feet (depending on the estimated depth of saturated sediments) below the ground surface using a steel plunger bar. After punching the plunger barhole, the plunger bar will be removed. Following removal, an appropriate length of polypropylene tubing that is either open or perforated on the bottom will be inserted into the temporary barhole and the barhole will be sealed at ground surface with low-permeability materials (bentonite or nearby surface soils). The portable gas detector will be attached to the end of the polypropylene tube and readings will be collected with the portable gas detector. The procedure for gathering readings from the temporary probe will be similar to the procedure described in Section 4.2. Once a stabilized reading is observed on the detector, it will be recorded.

5.0 RECORDKEEPING

This plan will be incorporated into the operating record along with the gas monitoring logs completed on a quarterly basis.

5.1 Quarterly Reporting

Quarterly methane concentrations and/or LEL for facility buildings and gas monitoring probes will be logged during each quarterly monitoring event. The manufacturer and model, as well as the status of the continuous monitor in each building will be recorded on the field data sheet. An example data sheet is provided in **Appendix B, Example Gas Migration Monitoring Field Forms**. Actual forms used may vary but must include date of monitoring and identification of monitoring personnel, weather/air temperature, make and model of portable gas meter, record of calibration, and percent methane or percent LEL at each monitoring point.

5.2 Plan Revisions

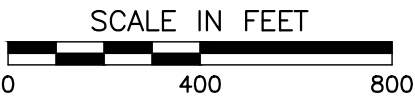
This plan should be revised and updated if it does not adequately protect the employees and public from methane. Revisions should also be made when there are substantial changes to the disposal area design or additional site buildings are constructed. Revisions should include any additional monitoring points necessary to meet the requirements of Rule 433.

FIGURES



LEGEND

- EXISTING PROPERTY LINE
- SOLID WASTE LIMITS
- SOLID WASTE CELL LIMITS
- BARHOLE LOCATION
- GAS PROBE LOCATION
- GAS DETECTION LOCATION
- PROPOSED GAS PROBE LOCATION



 Civil & Environmental Consultants, Inc. One SeaGate · Suite 2050 · Toledo, OH 43604 Ph: 419.724.5281 · Fax: 419.724.5286 www.cecinc.com		GAS MIGRATION MONITORING PLAN WEXFORD COUNTY LANDFILL WEXFORD COUNTY, MICHIGAN	
		SAMPLE LOCATION MAP	
DRAWN BY: RGS	CHECKED BY: CEP	APPROVED BY: SCP*	FIGURE NO.:
DATE: SEPTEMBER 2023	DWG SCALE: 1"=400'	PROJECT NO: 331-878	1

*HAND SIGNATURE ON FILE

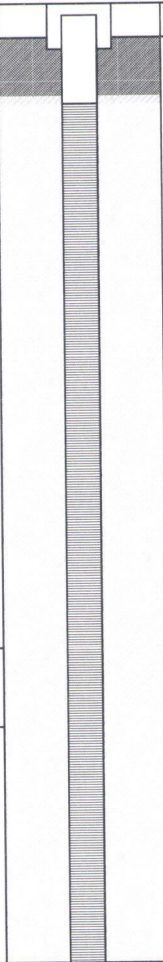
TABLES

TABLE 1.
GAS MIGRATION MONITORING SUMMARY
Wexford County Landfill - Gas Migration Monitoring Plan

Monitoring Point Identification	Date Installed	Coordinates		Ground Surface Elevation (ft AMSL)	Screened Interval (ft)	Bottom of Screen Elevation (ft AMSL)	Minimum 10 feet below floor	
		Northing	Easting				Waste Grade Elevation (ft AMSL)	(YES/NO)
Perimeter Gas Probes								
GP-1	12/29/2025	7492.30	8390.30	1398.15	5-70	1328.15	1380	YES
GP-2	12/29/2005	7446.40	9385.20	1391.58	5-70	1341.58	1380	YES
GP-3R	12/29/2005	7805.40	10075.00	1419.11	5-70	1349.11	1360	YES
GP-4	12/29/2005	8138.20	10127.70	1422.51	5-70	1352.51	1360	YES
GP-5R	4/11/2014	8786.94	10259.28	1419.45	5-80	1339.45	1360	YES
GP-6	12/29/2005	9350.70	10097.80	1420.00	5-115'	Decommissioned before Oct 2016		
GP-7	12/29/2005	10008.50	9184.60	1408.91	5-85	1323.91	1344	YES
GP-8	12/29/2005	9604.70	8571.40	1434.16	5-80	1354.16	1380/1330*	YES/NO*
GP-9	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GP-10	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GP-11	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GP-12	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GP-13	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GP-14	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GP-15	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Perimeter Barhole Locations								
BH-1	Located within future Cell I, south of MW-41							
BH-2	Located within future Cell 2, west of leachate tank							
BH-3	Located within future Cell 3, west of flare							
BH-4	Located within future Cell I, south of MW-9R							
Facility Structure Locations								
Scalehouse - Office		Located northwest of Cell I						
Maintenance Bldg.		Located west of Cell B						
Leachate Management Bldg.		Located west of Cell C						
Deep Well Bldg.		Located near Future Cell 4						
Notes:								
Ground surface elevation surveyed in October 2016.								
ft = feet								
AMSL = above mean sea level								
TBD = to be determined								
* = Based on current waste grades (1380 ft AMSL) the gas probe is a minimum of 10 feet below existing waste grade, however, proposed floor grades (1330 ft AMSL) for future cells, which results in less than 10 feet below waste grade.								



APPENDIX A
PERIMETER GAS PROBE LOGS

 CTI and Associates, Inc 12482 Emerson Drive Brighton, MI 48116 (248) 486-5100 phone (248) 486-5050 fax		GAS PROBE INSTALLATION LOG		GP-1		PAGE <u>1</u> of <u>1</u>	
PROJECT NAME: Wexford County Landfill				DATE: 12/26/05 - 12/29/05			
PROJECT NUMBER: 58043021				DEPTH TO WATER: N/A			
LOCATION: Cadillac, Michigan				TOTAL DEPTH: 70'			
LOGGED BY: F. Palatka				LOCATION: N7492.3, E8390.3			
DRILLING CONTRACTOR: Sheppler Drilling				DRILLER: N/A			
DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION		GAS PROBE DETAILS			
0		Ground Surface					
4		0.0' - 47.0' Brown SAND		 Bentonite Chips Fine Sand No. 30 Coarse Sand			
8							
12							
16							
20							
24							
28							
32							
36							
40							
44							
48		47.0' to 53.0' Brown CLAY					
52		53.0' Brown SAND					
56							
60							
64							
68							
72		End of Boring @70'					
76							
80							
84							
88							
92							
96							
100							

Well Materials: 2-inch Schedule
40 PVC w/ #10 slotted
perforations

Probes fitted with PVC cap
w/fitting and 4-inch steel casing
with locking cover



CTI and Associates, Inc
12482 Emerson Drive
Brighton, MI 48116
(248) 486-5100 phone
(248) 486-5050 fax

GAS PROBE INSTALLATION LOG

GP-2

PAGE 1 of 1

PROJECT NAME:	Wexford County Landfill	DATE:	12/26/05 - 12/29/05
PROJECT NUMBER:	58043021	DEPTH TO WATER:	N/A
LOCATION:	Cadillac, Michigan	TOTAL DEPTH:	50'
LOGGED BY:	F. Palatka	LOCATION:	N7446.4, E9385.2
DRILLING CONTRACTOR:	Sheppler Drilling	DRILLER:	N/A

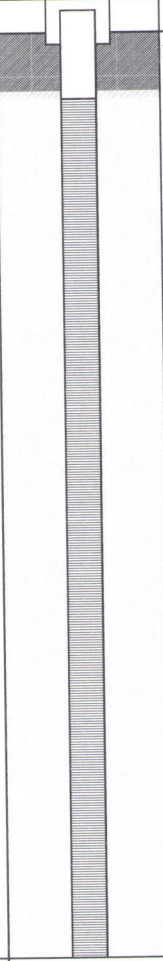
DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION	GAS PROBE DETAILS
0		Ground Surface	
2			Bentonite Chips
4			Fine Sand
6			No. 30 Coarse Sand
8			
10			
12			
14			
16			
18			
20			
22			
24			
26			
28			
30			
32			
34			
36			
38			
40			
42		42.0' - 48.0' Brown CLAY	Well Materials: 2-inch Schedule 40 PVC w/ #10 slotted perforations
44			
46			Probes fitted with PVC cap w/fitting and 4-inch steel casing with locking cover
48		48.0' - 50.0' Brown SAND	
50		End of Boring @ 50.0'	

 CTI and Associates, Inc 12482 Emerson Drive Brighton, MI 48116 (248) 486-5100 phone (248) 486-5050 fax			GAS PROBE INSTALLATION LOG		GP-3	PAGE <u>1</u> of <u>1</u>
PROJECT NAME: Wexford County Landfill				DATE: 12/26/05 - 12/29/05		
PROJECT NUMBER: 58043021				DEPTH TO WATER: N/A		
LOCATION: Cadillac, Michigan				TOTAL DEPTH: 70'		
LOGGED BY: F. Palatka				LOCATION: N7805.4, E10075.0		
DRILLING CONTRACTOR: Shepler Drilling				DRILLER: N/A		

DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION	GAS PROBE DETAILS	
0		Ground Surface		
4		0.0' - 47.0' Brown SAND	Bentonite Chips	
8			Fine Sand	
12			No. 30 Coarse Sand	
16				
20				
24				
28				
32				
36				
40			40.0' - 50.0' Gray silty SAND	
44				
48				
52				
56		50.0' - 70.0' Brown SAND		
60				
64				
68				
72		End of Boring @ 70'		
76				
80				
84				
88				
92				
96				
100				

Well Materials: 2-inch Schedule
 40 PVC w/ #10 slotted
 perforations

 Probes fitted with PVC cap
 w/fitting and 4-inch steel casing
 with locking cover

 CTI and Associates, Inc 12482 Emerson Drive Brighton, MI 48116 (248) 486-5100 phone (248) 486-5050 fax		GAS PROBE INSTALLATION LOG		GP-4		PAGE <u>1</u> of <u>1</u>	
PROJECT NAME: Wexford County Landfill				DATE: 12/26/05 - 12/29/05			
PROJECT NUMBER: 58043021				DEPTH TO WATER: N/A			
LOCATION: Cadillac, Michigan				TOTAL DEPTH: 70'			
LOGGED BY: F. Palatka				LOCATION: N8138.2, E10127.7			
DRILLING CONTRACTOR: Shepler Drilling				DRILLER: N/A			
DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION		GAS PROBE DETAILS			
0		Ground Surface					
4		0.0' - 9.0' Brown SAND					
8							
12		9.0' - 12.0' Brown CLAY					
16		12.0' - 70.0' Brown SAND					
20							
24							
28							
32							
36							
40							
44							
48							
52							
56							
60							
64							
68							
72		End of Boring @ 70'		Well Materials: 2-inch Schedule 40 PVC w/ #10 slotted perforations Probes fitted with PVC cap w/fitting and 4-inch steel casing with locking cover			
76							
80							
84							
88							
92							
96							
100							



CTI and Associates, Inc
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Brighton, MI 48116
(248) 486-5100 phone
(248) 486-5050 fax

GAS PROBE INSTALLATION LOG

GP-5

PAGE 1 of 1

PROJECT NAME:	Wexford County Landfill	DATE:	12/26/05 - 12/29/05
PROJECT NUMBER:	58043021	DEPTH TO WATER:	N/A
LOCATION:	Cadillac, Michigan	TOTAL DEPTH:	80'
LOGGED BY:	F. Palatka	LOCATION:	N8850.2, E10137.1
DRILLING CONTRACTOR:	Sheppler Drilling	DRILLER:	N/A

DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION	GAS PROBE DETAILS
0		Ground Surface	
4		0.0' - 60.0' Brown SAND	Bentonite Chips
8			Fine Sand
12			No. 30 Coarse Sand
16			
20			
24			
28			
32			
36			
40			
44			
48			
52			
56			
60			
64		60.0' - 63.0' Brown CLAY	
68		63.0' - 72.0' Brown SAND	
72		72.0' - 73.0' Brown CLAY	
76		73.0' - 80.0' Brown SAND	
80		End of Boring @ 80'	
84			Well Materials: 2-inch Schedule 40 PVC w/ #10 slotted perforations
88			
92			Probes fitted with PVC cap w/fitting and 4-inch steel casing with locking cover
96			
100			

LOG OF GP-5 abandonment

(Page 1 of 2)

PROJECT INFORMATION

Project Name : Wexford County Landfill, LLC
Project Number : 038
Project Location : Manton, Michigan
Project Manager : Daniel M. Staub

DRILLING INFORMATION

Date(s) Drilled : 04/11/14
Drilling Contractor : Shepler Well Drilling
Drill Rig : Mobile Drill
Drilling Method : HSA
Borehole Diameter : 4.25"
Field Log by : DMS
Electronic Log by : DMS

COMPLETION INFORMATION

Date Completed : 04/11/14
Mount Type : N/A
Well Diameter : 2.0"
Screened Interval : 5 - 80'
Screen Type : Sch. 40 PVC
Slot Size : 10
Cap Length : 0.5'
Joint Type : Flush-joint
Ground Elevation : approx. 1418.3

Depth (ft)	DESCRIPTION	USCS	GRAPHIC	SPLIT SPOON	BLOW COUNT	RECOVERY	PID (ppm)	Monitoring Well: GP-5 TOC Elevation: 1421.0
0	Soil not logged during well abandonment activities.							
5								
10								
15								
20								
25								
30								
35								
40								
45								

Bentonite Grout



Pescador, LLC

Notes: The drill rig was used to overdrill the 2-inch casing to 80 feet below surface grade utilizing 4.25-inch inner diameter augers. The drill rig then pulled/removed approximately 30 feet of the 2-inch PVC well. The remaining 50 feet of 2-inch PVC was destroyed in the process of overdrilling. A tremie pipe was used to grout the borehole from 80 feet to the surface grade.

LOG OF GP-5 abandonment

(Page 2 of 2)

PROJECT INFORMATION

Project Name : Wexford County Landfill, LLC
 Project Number : 038
 Project Location : Manton, Michigan
 Project Manager : Daniel M. Staub

DRILLING INFORMATION

Date(s) Drilled : 04/11/14
 Drilling Contractor : Shepler Well Drilling
 Drill Rig : Mobile Drill
 Drilling Method : HSA
 Borehole Diameter : 4.25"
 Field Log by : DMS
 Electronic Log by : DMS

COMPLETION INFORMATION

Date Completed : 04/11/14
 Mount Type : N/A
 Well Diameter : 2.0"
 Screened Interval : 5 - 80'
 Screen Type : Sch. 40 PVC
 Slot Size : 10
 Cap Length : 0.5'
 Joint Type : Flush-joint
 Ground Elevation : approx. 1418.3

Depth (ft)	DESCRIPTION	USCS	GRAPHIC	SPLIT SPOON	BLOW COUNT	RECOVERY	PID (ppm)	Monitoring Well: GP-5 TOC Elevation: 1421.0
45								
50								
55								
60								
65								
70								
75								
80	End of Boring at 80 feet.							
85								
90								

Bentonite Grout



Notes: The drill rig was used to overdrill the 2-inch casing to 80 feet below surface grade utilizing 4.25-inch inner diameter augers. The drill rig then pulled/removed approximately 30 feet of the 2-inch PVC well. The remaining 50 feet of 2-inch PVC was destroyed in the process of overdrilling. A tremie pipe was used to grout the borehole from 80 feet to the surface grade.

LOG OF GP-5 R

(Page 1 of 2)

PROJECT INFORMATION

Project Name : Wexford County Landfill, LLC
Project Number : 038
Project Location : Manton, Michigan
Project Manager : Daniel M. Staub

DRILLING INFORMATION

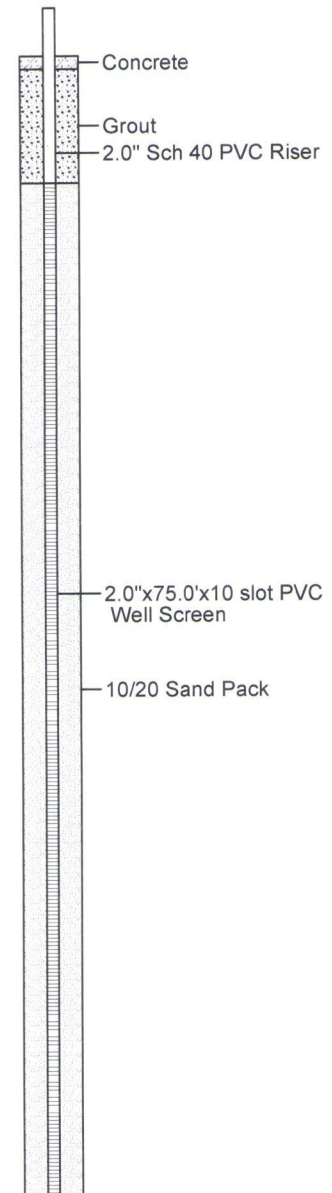
Date(s) Drilled : 04/11/13
Drilling Contractor : Shepler Well Drilling
Drill Rig : Mobile Drill
Drilling Method : HSA
Borehole Diameter : 4.25"
Field Log by : DMS
Electronic Log by : DMS

COMPLETION INFORMATION

Date Completed : 04/11/14
Mount Type : 4" x 4" Protective Cover
Well Diameter : 2.0"
Screened Interval : 5 - 80'
Screen Type : Sch. 40 PVC
Slot Size : 10
Cap Length : 3.5'
Joint Type : Flush-joint
Ground Elevation : 1419.00

Depth (ft)	DESCRIPTION	USCS	GRAPHIC	SPLIT SPOON	BLOW COUNT	RECOVERY	PID (ppm)
0	SAND, dark brown/black, fine, with organic/humus material, very moist.	SP					
5	SAND, fine-medium, few silt, orange, moist	SP					
10	SAND, fine-medium, few silt, yellow/tan, moist						
15							
20		SP					
25							
30							
35	SAND, fine-medium, few silt, tan/white/grey, moist						
40		SP					
45							

Monitoring Well: GP-5 R
TOC Elevation: 1421.98



LOG OF GP-5 R

(Page 2 of 2)

PROJECT INFORMATION

Project Name : Wexford County Landfill, LLC
Project Number : 038
Project Location : Manton, Michigan
Project Manager : Daniel M. Staub

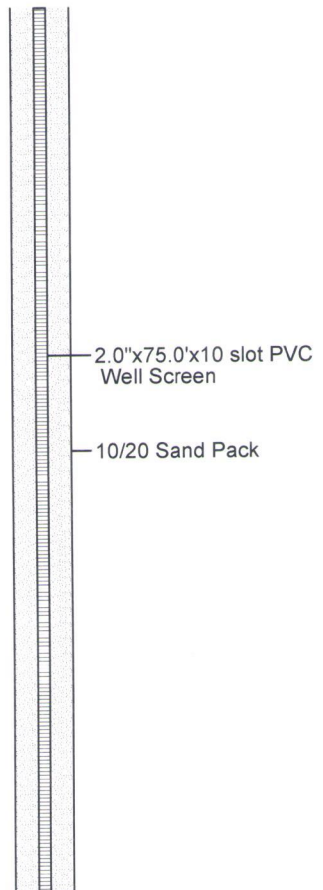
DRILLING INFORMATION

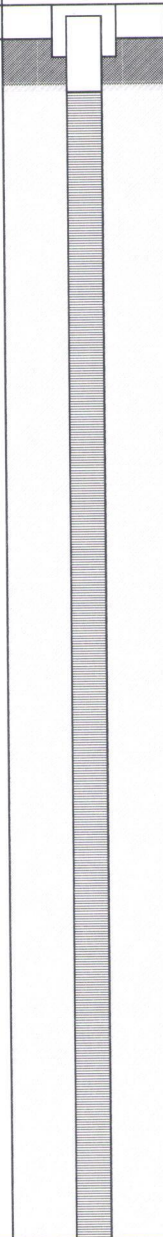
Date(s) Drilled : 04/11/13
Drilling Contractor : Shepler Well Drilling
Drill Rig : Mobile Drill
Drilling Method : HSA
Borehole Diameter : 4.25"
Field Log by : DMS
Electronic Log by : DMS

COMPLETION INFORMATION

Date Completed : 04/11/14
Mount Type : 4" x 4" Protective Cover
Well Diameter : 2.0"
Screened Interval : 5 - 80'
Screen Type : Sch. 40 PVC
Slot Size : 10
Cap Length : 3.5'
Joint Type : Flush-joint
Ground Elevation : 1419.00

Depth (ft)	DESCRIPTION	USCS	GRAPHIC	SPLIT SPOON	BLOW COUNT	RECOVERY	PID (ppm)	Monitoring Well: GP-5 R TOC Elevation: 1421.98
45		SP						
50	SANDY SILT, fine, some clay, tan, saturated at 65'							
55		SM						
60								
65								
70	CLAYEY SAND, fine-medium, some silt, tan, moist	CL						
75	SANDY CLAY, fine-medium, some silt, tan/grey, moist	SC						
80	End of Boring at 80 feet							
85								
90								



 CTT and Associates, Inc 12482 Emerson Drive Brighton, MI 48116 (248) 486-5100 phone (248) 486-5050 fax		GAS PROBE INSTALLATION LOG		GP-6		PAGE <u>1</u> of <u>1</u>	
PROJECT NAME: Wexford County Landfill				DATE: 12/26/05 - 12/29/05			
PROJECT NUMBER: 58043021				DEPTH TO WATER: N/A			
LOCATION: Cadillac, Michigan				TOTAL DEPTH: 115'			
LOGGED BY: F. Palatka				LOCATION: N9350.7, E10097.8			
DRILLING CONTRACTOR: Sheppler Drilling				DRILLER: N/A			
DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION		GAS PROBE DETAILS			
0		Ground Surface					
5		0.0' - 115.0' Brown SAND		 Bentonite Chips Fine Sand No. 30 Coarse Sand			
10							
15							
20							
25							
30							
35							
40							
45							
50							
55							
60							
65							
70							
75							
80							
85							
90							
95							
100							
105							
110							
115							
		End of Boring @ 115'		Well Materials: 2-inch Schedule 40 PVC w/ #10 slotted perforations Probes fitted with PVC cap w/fitting and 4-inch steel casing with locking cover			
120							
125							



CTI and Associates, Inc
12482 Emerson Drive
Brighton, MI 48116
(248) 486-5100 phone
(248) 486-5050 fax

GAS PROBE INSTALLATION LOG

GP-7

PAGE 1 of 1

PROJECT NAME:	Wexford County Landfill	DATE:	12/26/05 - 12/29/05
PROJECT NUMBER:	58043021	DEPTH TO WATER:	N/A
LOCATION:	Cadillac, Michigan	TOTAL DEPTH:	85'
LOGGED BY:	F. Palatka	LOCATION:	N10008.5, E9184.6
DRILLING CONTRACTOR:	Shepler Drilling	DRILLER:	N/A

DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION	GAS PROBE DETAILS
0		Ground Surface	
4		0.0' - 85.0' Brown SAND	Bentonite Chips
8			Fine Sand
12			No. 30 Coarse Sand
16			
20			
24			
28			
32			
36			
40			
44			
48			
52			
56			
60			
64			Well Materials: 2-inch Schedule 40 PVC w/ #10 slotted perforations
68			Probes fitted with PVC cap w/fitting and 4-inch steel casing with locking cover
72			
76			
80			
84			
88		End of Boring @ 85'	
92			
96			
100			



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GAS PROBE INSTALLATION LOG

GP-8

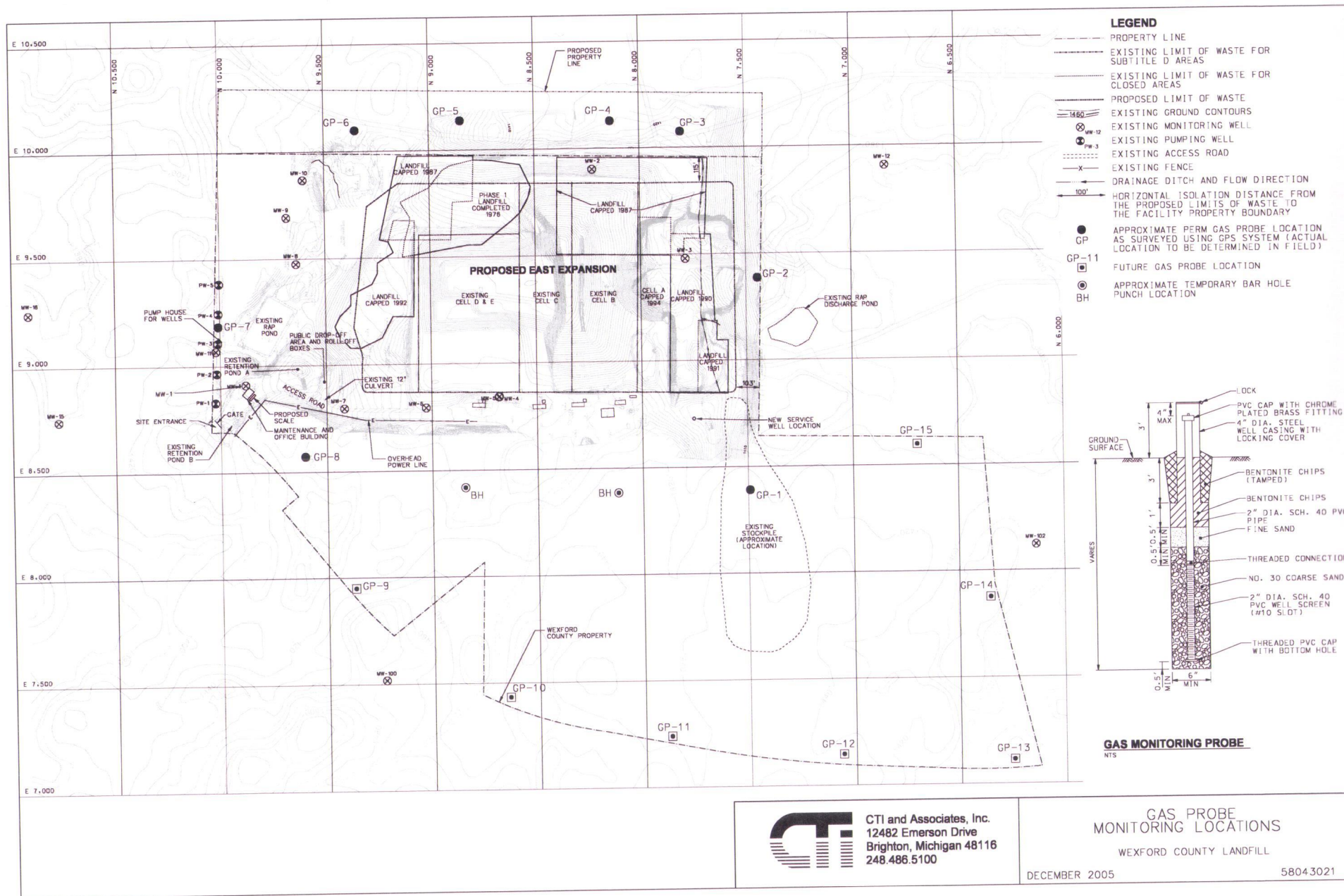
PAGE 1 of 1

PROJECT NAME:	Wexford County Landfill	DATE:	12/26/05 - 12/29/05
PROJECT NUMBER:	58043021	DEPTH TO WATER:	N/A
LOCATION:	Cadillac, Michigan	TOTAL DEPTH:	80'
LOGGED BY:	F. Palatka	LOCATION	N9604.7, E8571.4
DRILLING CONTRACTOR:	Sheppler Drilling	DRILLER:	N/A

DEPTH IN FEET	GRAPHIC LOG	LITHOLOGICAL DESCRIPTION	GAS PROBE DETAILS
0		Ground Surface	
4		0.0' - 40.0' Brown SAND	Bentonite Chips Fine Sand No. 30 Coarse Sand
8			
12			
16			
20			
24			
28			
32			
36			
40			
44		40.0' - 41.0' Red CLAY	
48		41.0' - 68.0' Brown SAND	
52			
56			
60			
64			
68		68.0' - 69.0' Red CLAY	
72		69.0' - 80.0' Brown SAND	
76			
80		End of Boring @ 80'	
84			
88			
92			
96			
100			

Well Materials: 2-inch Schedule
40 PVC w/ #10 slotted
perforations

Probes fitted with PVC cap
w/fitting and 4-inch steel casing
with locking cover



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Brighton, Michigan 48116
248.486.5100

APPENDIX B
EXAMPLE GAS MIGRATION MONITORING FIELD FORMS

PERIMETER LANDFILL GAS MONITORING FORM
GAS MIGRATION MONITORING PLAN

Monitored by: _____
Date: _____ Temperature: _____
Equipment: _____ Date of Calibration: _____

Permanent LFG Probes				
Monitoring Point Identification	Percent Methane by Volume (0-5%)	Percentage of the Lower Explosive Limit (LEL) (0-100%)	Percent Oxygen (O ₂) by Volume	Comments
GP-1				
GP-2				
GP-3				
GP-4				
GP-5R				
GP-7				
GP-8				
Notes:				

PERIMETER LANDFILL GAS MONITORING FORM

GAS MIGRATION MONITORING PLAN

Monitored by: _____

Date: _____ Temperature: _____

Equipment: _____ Date of Calibration: _____

Ambient Air and Barhole Monitoring Points							
Monitoring Point Identification	Ambient Air			Barhole			Comments
	% Methane by Volume (0-5%)	% LEL (0-100%)	Percent O ₂ by Volume	% Methane by Volume (0-5%)	% LEL (0-100%)	Percent O ₂ by Volume	
BH-1							
BH-2							
BH-3							
BH-4							

Notes:

PERIMETER LANDFILL GAS MONITORING FORM

GAS MIGRATION MONITORING PLAN

Monitored by: _____
 Date: _____ Air Temperature: _____
 Equipment: _____ Date of Calibration: _____

Electronic LFG Detectors							
Monitoring Point Identification	Permanent Detector (Make and Model)	Visual Inspection	Alarm Test	Ambient Air			Comments
				% Methane by Volume (0-5%)	% LEL (0-100%)	Percent O ₂ by Volume	
Scalehouse/Office							
Maintenance Bldg.							
Leachate Management Bldg.							
Deep Well Bldg.							

Notes:

HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA
Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
GP-1	1/24/2018	9:30AM-12:30PM	0.10	2.00	0.40	22.70	76.70	
GP-1	5/21/2018	8:00AM-10:30AM	0.10	2.00	1.10	19.90	78.80	
GP-1	8/22/2018	7:30AM-10:30AM	0.00	0.00	1.60	19.70	78.60	8/22 & 8/23 on field form
GP-1	10/22/2018	10:00AM-4:40PM	0.00	0.00	1.80	20.10	78.00	
GP-1	2/21/2019	9:00AM-11:30AM	0.00	0.00	2.00	21.30	76.60	
GP-1	6/14/2019	9:00AM-12:30PM	0.00	0.00	1.60	18.60	79.60	
GP-1	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.90	20.10	78.90	
GP-1	12/10/2019	9:00AM-12:30PM	0.00	0.00	1.10	20.50	78.20	
GP-1	3/17/2020	10:45AM-2:00PM	0.00	0.00	1.10	21.50	77.30	
GP-1	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.10	20.80	78.40	
GP-1	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.90	19.50	79.60	
GP-1	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.30	21.90	77.80	
GP-1	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.90	22.80	76.30	
GP-1	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.60	21.30	78.10	
GP-1	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.80	19.50	79.70	
GP-1	10/29/2021	10:30AM-2:30PM	0.00	0.00	1.40	18.20	80.40	
GP-1	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.50	23.30	76.10	
GP-1	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.50	19.60	79.90	
GP-1	8/25/2022	9:20AM-12:20PM	0.00	0.00	1.40	19.10	79.50	
GP-1	10/12/2022	9:45AM-12:30PM	0.10	2.00	1.50	18.80	79.60	
GP-1	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.70	21.30	78.00	
GP-1	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.90	18.60	80.40	
GP-1	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.10	20.90	79.00	
GP-2	1/24/2018	9:30AM-12:30PM	0.10	2.00	3.20	21.50	75.10	
GP-2	5/21/2018	8:00AM-10:30AM	0.10	2.00	3.70	18.20	77.90	
GP-2	8/22/2018	7:30AM-10:30AM	0.00	0.00	5.00	17.30	77.70	8/22 & 8/23 on field form
GP-2	10/22/2018	10:00AM-4:40PM	0.00	0.00	6.30	16.60	76.80	
GP-2	2/21/2019	9:00AM-11:30AM	0.00	0.00	5.30	18.80	75.60	
GP-2	6/14/2019	9:00AM-12:30PM	0.00	0.00	3.40	17.30	79.20	
GP-2	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.30	18.00	79.00	
GP-2	12/10/2019	9:00AM-12:30PM	0.00	0.00	3.80	22.00	74.10	
GP-2	3/17/2020	10:45AM-2:00PM	0.00	0.00	3.10	19.60	77.30	
GP-2	6/12/2020	10:45AM-2:00PM	0.00	0.00	1.70	21.10	77.20	
GP-2	7/16/2020	9:20AM-12:20PM	0.00	0.00	2.20	18.70	79.10	
GP-2	10/20/2020	10:00AM-1:00PM	0.00	0.00	1.40	20.80	77.70	
GP-2	1/22/2021	8:30AM-1:15PM	0.00	0.00	2.40	21.10	76.50	
GP-2	4/21/2021	12:00PM-2:30PM	0.10	2.00	1.60	20.60	77.70	
GP-2	7/21/2021	12:00PM-3:20PM	0.00	0.00	2.40	17.90	79.70	
GP-2	10/29/2021	10:30AM-2:30PM	0.00	0.00	6.50	13.20	80.30	
GP-2	3/10/2022	1:00PM-4:00PM	0.00	0.00	2.00	21.90	76.20	
GP-2	5/12/2022	12:30PM-3:30PM	0.00	0.00	1.70	18.50	79.80	
GP-2	8/25/2022	9:20AM-12:20PM	0.00	0.00	3.50	17.80	78.70	
GP-2	10/12/2022	9:45AM-12:30PM	0.10	2.00	6.00	15.10	78.80	
GP-2	2/8/2023	8:00AM-11:00AM	0.00	0.00	2.40	20.20	77.40	
GP-2	4/13/2023	12:00PM-2:00PM	0.00	0.00	4.80	15.20	80.00	
GP-2	8/1/2023	7:00AM-9:30AM	0.00	0.00	2.20	19.20	78.70	

1 - Values were recorded on the quarterly field forms.
2 - Values were calculated based on values recorded on the quarterly field form.

HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA

Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
GP-3R	1/24/2018	9:30AM-12:30PM	0.10	2.00	0.20	21.20	78.60	
GP-3R	5/21/2018	8:00AM-10:30AM	0.10	2.00	0.20	20.70	79.00	
GP-3R	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.50	78.30	8/22 & 8/23 on field form
GP-3R	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.10	22.20	77.60	
GP-3R	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.20	20.70	79.10	
GP-3R	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.10	20.60	79.10	
GP-3R	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.00	21.20	78.70	
GP-3R	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.20	20.30	79.40	
GP-3R	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.10	22.10	77.70	
GP-3R	6/12/2020	10:45AM-2:00PM	0.10	2.00	0.10	23.00	76.80	
GP-3R	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.10	20.70	79.30	
GP-3R	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.10	21.60	78.20	
GP-3R	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.10	22.00	77.90	
GP-3R	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.10	21.40	78.40	
GP-3R	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.10	20.20	79.80	
GP-3R	10/29/2021	10:30AM-2:30PM	0.00	0.00	0.10	20.60	79.30	
GP-3R	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.10	22.90	77.00	
GP-3R	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.00	20.40	79.60	
GP-3R	8/25/2022	9:20AM-12:20PM	0.00	0.00	1.50	19.40	79.00	
GP-3R	10/12/2022	9:45AM-12:30PM	0.10	2.00	1.60	19.10	79.20	
GP-3R	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.30	22.00	77.70	
GP-3R	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.90	20.00	74.20	
GP-3R	8/1/2023	7:00AM-9:30AM	0.00	0.00	1.60	19.30	79.20	
GP-4R	1/24/2018	9:30AM-12:30PM	0.00	0.00	0.10	21.90	76.80	well has been extended 15 feet
GP-4R	5/21/2018	8:00AM-10:30AM	0.10	2.00	0.20	20.60	79.00	well has been extended 15 feet
GP-4R	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.40	78.40	8/22 & 8/23 on field form, well has been extended 15 feet
GP-4R	10/22/2018	10:00AM-4:40PM	0.60	12.00	6.50	14.20	78.70	well has been extended 15 feet
GP-4R	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.10	20.60	79.20	well has been extended 15 feet
GP-4R	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.10	20.60	79.20	well has been extended 15 feet
GP-4R	9/10/2019	9:00AM-12:30PM	0.00	0.00	12.90	4.70	82.30	well has been extended 15 feet
GP-4R	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.20	20.60	79.80	well has been extended 15 feet
GP-4R	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.10	21.90	77.90	well has been extended 15 feet
GP-4R	6/12/2020	10:45AM-2:00PM	0.10	2.00	0.10	22.10	77.80	well has been extended 15 feet
GP-4R	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.10	20.70	79.30	well has been extended 15 feet
GP-4R	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.10	21.60	78.20	well has been extended 15 feet
GP-4R	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.10	21.80	78.10	well has been extended 15 feet
GP-4R	4/21/2021	12:00PM-2:30PM	0.20	4.00	0.10	21.30	78.50	well has been extended 15 feet
GP-4R	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.10	20.20	79.80	well has been extended 15 feet
GP-4R	10/29/2021	10:30AM-2:30PM	24.50	490.00	26.70	0.10	48.70	well has been extended 15 feet, stepped out with three barholes (GP4R BH-A, GP4R BH-B, and GP-4R BH-C)
GP-4R	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.10	22.60	77.30	well has been extended 15 feet
GP-4R	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.20	20.00	79.90	well has been extended 15 feet
GP-4R	8/25/2022	9:20AM-12:20PM	0.00	0.00	3.30	16.30	20.40	well has been extended 15 feet
GP-4R	10/12/2022	9:45AM-12:30PM	11.70	234.00	26.70	0.00	61.90	well has been extended 15 feet, gas extraction system and flares were off for repair, conducted verification monitoring on October 19 and 20.
GP-4R	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.10	21.90	78.00	well has been extended 15 feet
GP-4R	4/13/2023	12:00PM-2:00PM	0.00	0.00	10.20	10.30	79.50	well has been extended 15 feet
GP-4R	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.10	20.70	79.20	well has been extended 15 feet
GP-4R BH-A	10/29/2021	10:30AM-2:30PM	0.00	0.00	2.00	17.10	80.90	25 feet east of GP-4R, approximately 0-3 feet below ground surface
GP-4R BH-B	10/29/2021	10:30AM-2:30PM	0.00	0.00	1.00	16.30	82.70	25 feet north of BH-A, approximately 0-3 feet below ground surface
GP-4R BH-C	10/29/2021	10:30AM-2:30PM	0.00	0.00	0.70	20.10	79.20	25 feet south of BH-A, approximately 0-3 feet below ground surface

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HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA
Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
GP-5R	1/24/2018	9:30AM-12:30PM	0.00	0.00	1.30	22.40	76.10	
GP-5R	5/21/2018	8:00AM-10:30AM	0.10	2.00	3.00	19.20	77.60	
GP-5R	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.90	20.90	78.20	8/22 & 8/23 on field form
GP-5R	10/22/2018	10:00AM-4:40PM	0.00	0.00	5.30	16.10	78.50	
GP-5R	2/21/2019	9:00AM-11:30AM	0.00	0.00	5.00	17.10	77.70	
GP-5R	6/14/2019	9:00AM-12:30PM	0.00	0.00	1.60	18.80	79.50	
GP-5R	9/10/2019	9:00AM-12:30PM	0.00	0.00	6.00	13.10	80.80	
GP-5R	12/10/2019	9:00AM-12:30PM	0.10	2.00	8.00	16.60	75.20	
GP-5R	3/17/2020	10:45AM-2:00PM	0.70	14.00	9.70	10.30	79.10	
GP-5R	6/12/2020	10:45AM-2:00PM	0.00	0.00	1.10	21.30	77.60	
GP-5R	7/16/2020	9:20AM-12:20PM	0.00	0.00	7.20	11.90	80.90	
GP-5R	10/20/2020	10:00AM-1:00PM	0.00	0.00	2.20	19.80	78.00	
GP-5R	1/22/2021	8:30AM-1:15PM	0.00	0.00	8.50	18.50	73.00	
GP-5R	4/21/2021	12:00PM-2:30PM	0.20	4.00	3.10	19.70	77.00	
GP-5R	7/21/2021	12:00PM-3:20PM	0.00	0.00	5.70	16.20	78.10	
GP-5R	10/29/2021	10:30AM-2:30PM	2.90	58.00	19.30	0.20	77.60	
GP-5R	3/10/2022	1:00PM-4:00PM	0.00	0.00	5.30	20.40	74.40	
GP-5R	5/12/2022	12:30PM-3:30PM	0.00	0.00	5.10	16.70	78.20	
GP-5R	8/25/2022	9:20AM-12:20PM	0.10	2.00	17.60	3.90	78.40	
GP-5R	10/12/2022	9:45AM-12:30PM	5.80	116.00	22.00	0.00	72.60	gas extraction system and flares were off for repair, conducted verification monitoring on the 19th and 20th.
GP-5R	2/8/2023	8:00AM-11:00AM	0.00	0.00	8.30	18.90	72.90	
GP-5R	4/13/2023	12:00PM-2:00PM	3.20	64.00	19.80	0.30	76.80	
GP-5R	8/1/2023	7:00AM-9:30AM	0.00	0.00	10.60	12.00	77.40	
GP-6	1/24/2018	9:30AM-12:30PM	0.10	2.00	0.10	22.60	77.00	roughly 20 feet of the well has been removed
GP-6	5/21/2018	8:00AM-10:30AM	NA	NA	NA	NA	NA	well has been permanently removed
GP-6	8/22/2018	7:30AM-10:30AM	NA	NA	NA	NA	NA	8/22 & 8/23 on field form, well removed
GP-6	10/22/2018	10:00AM-4:40PM	NA	NA	NA	NA	NA	well removed
GP-6	2/21/2019	9:00AM-11:30AM	NA	NA	NA	NA	NA	well removed
GP-6	6/14/2019	9:00AM-12:30PM	NA	NA	NA	NA	NA	well removed
GP-6	9/10/2019	9:00AM-12:30PM	NA	NA	NA	NA	NA	well removed
GP-6	12/10/2019	9:00AM-12:30PM	NA	NA	NA	NA	NA	well removed
GP-6	3/17/2020	10:45AM-2:00PM	NA	NA	NA	NA	NA	well removed
GP-6	6/12/2020	10:45AM-2:00PM	NA	NA	NA	NA	NA	well removed
GP-6	7/16/2020	9:20AM-12:20PM	NA	NA	NA	NA	NA	well removed
GP-6	10/20/2020	10:00AM-1:00PM	NA	NA	NA	NA	NA	well removed
GP-6	1/22/2021	8:30AM-1:15PM	NA	NA	NA	NA	NA	well removed
GP-6	4/21/2021	12:00PM-2:30PM	NA	NA	NA	NA	NA	well removed
GP-6	7/21/2021	12:00PM-3:20PM	NA	NA	NA	NA	NA	well removed
GP-6	10/29/2021	10:30AM-2:30PM	NA	NA	NA	NA	NA	well removed
GP-6	3/10/2022	1:00PM-4:00PM	NA	NA	NA	NA	NA	well removed
GP-6	5/12/2022	12:30PM-3:30PM	NA	NA	NA	NA	NA	well removed
GP-6	8/25/2022	9:20AM-12:20PM	NA	NA	NA	NA	NA	well removed
GP-6	10/12/2022	9:45AM-12:30PM	NA	NA	NA	NA	NA	well removed
GP-6	2/8/2023	8:00AM-11:00AM	NA	NA	NA	NA	NA	well removed
GP-6	4/13/2023	12:00PM-2:00PM	NA	NA	NA	NA	NA	well removed
GP-6	8/1/2023	7:00AM-9:30AM	NA	NA	NA	NA	NA	well removed

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HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA
Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
GP-7	1/24/2018	9:30AM-12:30PM	0.00	0.00	0.30	21.90	77.60	
GP-7	5/21/2018	8:00AM-10:30AM	0.10	2.00	0.8	19.9	79.10	
GP-7	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.20	21.00	78.30	8/22 & 8/23 on field form
GP-7	10/22/2018	10:00AM-4:40PM	0.00	0.00	1.20	20.60	78.00	
GP-7	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.90	20.60	78.40	
GP-7	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.80	18.90	19.10	balance appears anomalous
GP-7	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.70	20.50	78.80	
GP-7	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.80	22.00	77.10	
GP-7	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.70	20.40	78.80	
GP-7	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.40	21.20	78.40	
GP-7	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.70	19.50	79.70	
GP-7	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.30	20.80	78.80	
GP-7	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.60	21.00	78.30	
GP-7	4/21/2021	12:00PM-2:30PM	0.20	4.00	0.40	20.50	78.90	
GP-7	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.90	19.40	79.70	
GP-7	10/29/2021	10:30AM-2:30PM	0.10	2.00	1.30	17.30	81.30	
GP-7	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.40	21.70	77.80	
GP-7	5/12/2022	12:30PM-3:30PM	0.10	2.00	0.40	19.70	79.80	
GP-7	8/25/2022	9:20AM-12:20PM	0.00	0.00	1.60	19.00	79.80	
GP-7	10/12/2022	9:45AM-12:30PM	0.10	2.00	1.40	18.90	79.70	
GP-7	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.50	21.10	78.40	
GP-7	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.80	18.30	80.90	
GP-7	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.60	19.80	79.60	
GP-8	1/24/2018	9:30AM-12:30PM	0.10	2.00	1.00	22.30	76.60	
GP-8	5/21/2018	8:00AM-10:30AM	0.00	0.00	1.1	20.2	78.50	
GP-8	8/22/2018	7:30AM-10:30AM	0.00	0.00	1.30	20.20	78.40	8/22 & 8/23 on field form
GP-8	10/22/2018	10:00AM-4:40PM	0.00	0.00	1.80	20.40	77.70	
GP-8	2/21/2019	9:00AM-11:30AM	0.00	0.00	1.80	20.40	77.70	
GP-8	6/14/2019	9:00AM-12:30PM	0.10	2.00	1.40	19.50	79.00	
GP-8	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.90	19.90	79.10	
GP-8	12/10/2019	9:00AM-12:30PM	0.00	0.00	1.60	22.50	75.70	
GP-8	3/17/2020	10:45AM-2:00PM	0.00	0.00	1.20	21.30	77.40	
GP-8	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.80	20.10	79.00	
GP-8	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.90	19.10	79.90	
GP-8	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.60	21.90	77.50	
GP-8	1/22/2021	8:30AM-1:15PM	0.00	0.00	1.30	21.60	77.10	
GP-8	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.90	20.90	78.10	
GP-8	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.90	19.10	80.00	
GP-8	10/29/2021	10:30AM-2:30PM	0.00	0.00	1.60	19.20	79.20	
GP-8	3/10/2022	1:00PM-4:00PM	0.00	0.00	1.00	22.40	76.60	
GP-8	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.70	20.10	79.30	
GP-8	8/25/2022	9:20AM-12:20PM	0.00	0.00	1.30	19.70	79.30	
GP-8	10/12/2022	9:45AM-12:30PM	0.10	2.00	1.50	19.90	78.50	
GP-8	2/8/2023	8:00AM-11:00AM	0.00	0.00	1.10	21.50	77.40	
GP-8	4/13/2023	12:00PM-2:00PM	0.00	0.00	1.10	18.90	80.00	
GP-8	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.90	20.10	79.00	

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HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA

Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
BH-1	1/24/2018	9:30AM-12:30PM	0.00	0.00	0.10	22.40	77.30	
BH-1	5/21/2018	8:00AM-10:30AM	0.10	2.00	0.5	20.3	79	
BH-1	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.30	21.2	78.50	8/22 & 8/23 on field form
BH-1	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.80	21.50	77.60	vicinity of the NW corner of new cell
BH-1	2/21/2019	9:00AM-11:30AM	NA	NA	NA	NA	NA	unable to access due depth of snow, vicinity of the NW corner of new cell
BH-1	6/14/2019	9:00AM-12:30PM	0.00	0.00	1.00	19.50	79.40	vicinity of the NW corner of new cell
BH-1	9/10/2019	9:00AM-12:30PM	NA	NA	NA	NA	NA	vicinity of the NW corner of new cell, no note regarding why NA
BH-1	12/10/2019	9:00AM-12:30PM	NA	NA	NA	NA	NA	vicinity of the NW corner of new cell, no note regarding why NA
BH-1	3/17/2020	10:45AM-2:00PM	NA	NA	NA	NA	NA	vicinity of the NW corner of new cell, no note regarding why NA
BH-1	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.30	22.50	77.10	vicinity of the NW corner of new cell
BH-1	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.50	19.90	79.50	vicinity of the NW corner of new cell
BH-1	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.60	21.80	77.60	vicinity of the NW corner of new cell
BH-1	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.60	21.80	77.60	vicinity of the NW corner of new cell
BH-1	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.60	21.80	78.50	vicinity of the NW corner of new cell
BH-1	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.60	21.80	78.50	vicinity of the NW corner of new cell
BH-1	10/29/2021	10:30AM-2:30PM	0.00	0.00	0.60	21.80	78.50	vicinity of the NW corner of new cell
BH-1	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.60	21.80	78.50	centrally located on the north side of Cell H
BH-1	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.10	20.00	79.80	centrally located on the north side of Cell H
BH-1	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.10	20.00	79.80	centrally located on the north side of Cell H
BH-1	10/12/2022	9:45AM-12:30PM	0.10	2.00	0.20	21.20	78.40	centrally located on the north side of Cell H
BH-1	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.10	22.00	77.90	centrally located on the north side of Cell H
BH-1	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.00	20.10	79.80	centrally located on the north side of Cell H
BH-1	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.00	20.10	79.80	centrally located on the north side of Cell H
BH-2	1/24/2018	9:30AM-12:30PM	0.00	0.00	0.10	22.40	77.30	
BH-2	5/21/2018	8:00AM-10:30AM	0.00	0.00	2.7	18.6	78.6	
BH-2	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.70	20.6	78.60	8/22 & 8/23 on field form
BH-2	10/22/2018	10:00AM-4:40PM	0.00	0.00	1.20	21.20	77.50	SW of lagoon
BH-2	2/21/2019	9:00AM-11:30AM	NA	NA	NA	NA	NA	unable to access due depth of snow, SW of lagoon
BH-2	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.10	19.20	80.00	SW of lagoon
BH-2	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.40	20.70	78.90	SW of lagoon
BH-2	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.40	22.60	76.80	SW of lagoon
BH-2	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.30	22.80	76.80	SW of lagoon
BH-2	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.90	19.80	79.30	SW of lagoon
BH-2	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.50	18.70	80.70	SW of lagoon
BH-2	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.50	22.00	77.50	SW of lagoon
BH-2	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.50	22.00	77.50	SW of lagoon
BH-2	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.50	21.30	78.50	SW of lagoon
BH-2	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.50	21.30	78.50	SW of lagoon
BH-2	10/29/2021	10:30AM-2:30PM	0.00	0.00	0.20	20.90	78.90	SW of lagoon
BH-2	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.40	23.30	76.30	SW of lagoon
BH-2	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.20	20.70	79.20	SW of lagoon
BH-2	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.40	20.60	79.20	SW of lagoon
BH-2	10/12/2022	9:45AM-12:30PM	0.10	2.00	0.30	21.20	78.40	SW of lagoon
BH-2	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.20	21.50	78.30	SW of lagoon
BH-2	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.00	20.10	79.80	SW of lagoon
BH-2	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.30	20.70	79.10	SW of lagoon

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Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
BH-3	1/24/2018	9:30AM-12:30PM	0.00	0.00	0.10	22.40	76.90	
BH-3	5/21/2018	8:00AM-10:30AM	0.10	2.00	1.1	19.8	79	
BH-3	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.20	21	78.70	8/22 & 8/23 on field form
BH-3	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.80	21.90	77.20	NW of lagoon
BH-3	2/21/2019	9:00AM-11:30AM	NA	NA	NA	NA	NA	unable to access due depth of snow, NW of lagoon
BH-3	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.60	19.50	79.70	NW of lagoon
BH-3	9/10/2019	9:00AM-12:30PM	0.00	0.00	1.80	19.30	78.80	NW of lagoon
BH-3	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.50	22.50	76.90	NW of lagoon
BH-3	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.30	22.50	77.10	NW of lagoon
BH-3	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.80	20.10	79.00	NW of lagoon
BH-3	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.80	20.10	79.00	NW of lagoon
BH-3	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.50	21.70	77.80	NW of lagoon
BH-3	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.50	21.70	77.80	NW of lagoon
BH-3	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.50	21.40	78.40	NW of lagoon
BH-3	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.50	21.40	78.40	NW of lagoon
BH-3	10/29/2021	10:30AM-2:30PM	0.00	0.00	0.30	20.90	78.90	NW of lagoon
BH-3	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.20	23.70	76.10	NW of lagoon
BH-3	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.20	20.70	79.20	NW of lagoon
BH-3	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.50	20.50	79.20	NW of lagoon
BH-3	10/12/2022	9:45AM-12:30PM	0.10	2.00	0.20	21.30	78.40	NW of lagoon
BH-3	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.20	21.80	78.00	NW of lagoon
BH-3	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.00	20.10	79.80	NW of lagoon
BH-3	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.30	20.70	79.00	NW of lagoon
BH-4	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.10	22.1	77.70	bore hole punch at toe of slope on the north side
BH-4	2/21/2019	9:00AM-11:30AM	NA	NA	NA	NA	NA	unable to access due depth of snow, bore hole punch at toe of slope on the north side
BH-4	6/14/2019	9:00AM-12:30PM	0.10	2.00	0.10	20.90	78.90	bore hole punch at toe of slope on the north side
BH-4	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.20	21.10	78.60	bore hole punch at toe of slope on the north side
BH-4	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.20	19.40	80.20	bore hole punch at toe of slope on the north side
BH-4	3/17/2020	10:45AM-2:00PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	6/12/2020	10:45AM-2:00PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	7/16/2020	9:20AM-12:20PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	10/20/2020	10:00AM-1:00PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	1/22/2021	8:30AM-1:15PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	4/21/2021	12:00PM-2:30PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	7/21/2021	12:00PM-3:20PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	10/29/2021	10:30AM-2:30PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	3/10/2022	1:00PM-4:00PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	5/12/2022	12:30PM-3:30PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	8/25/2022	9:20AM-12:20PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	10/12/2022	9:45AM-12:30PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	2/8/2023	8:00AM-11:00AM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	4/13/2023	12:00PM-2:00PM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole
BH-4	8/1/2023	7:00AM-9:30AM	NA	NA	NA	NA	NA	bore hole punch at toe of slope on the north side, water in borehole

1 - Values were recorded on the quarterly field forms.
2 - Values were calculated based on values recorded on the quarterly field form.

HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA

Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
Scalehouse-Office	1/24/2018	9:30AM-12:30PM	0.00	0.00	0.20	20.70	79.10	
Scalehouse-Office	5/21/2018	8:00AM-10:30AM	0.20	4.00	0.10	20.70	78.80	
Scalehouse-Office	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.20	48.70	8/22 & 8/23 on field form, balance appears to be anomalous
Scalehouse-Office	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.20	22.00	77.70	
Scalehouse-Office	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.20	22.10	77.60	
Scalehouse-Office	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.20	19.90	79.80	
Scalehouse-Office	9/10/2019	9:00AM-12:30PM	NA	NA	NA	NA	NA	no note regarding why NA
Scalehouse-Office	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.30	21.90	77.80	
Scalehouse-Office	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.30	21.90	77.80	
Scalehouse-Office	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.10	21.60	78.30	
Scalehouse-Office	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.10	20.70	79.20	
Scalehouse-Office	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.20	20.90	78.90	
Scalehouse-Office	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.10	20.60	79.20	
Scalehouse-Office	4/21/2021	12:00PM-2:30PM	0.20	4.00	0.10	20.70	79.00	
Scalehouse-Office	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.10	20.50	79.40	
Scalehouse-Office	10/29/2021	10:30AM-2:30PM	0.10	2.00	0.20	20.00	79.70	
Scalehouse-Office	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.20	21.50	78.30	
Scalehouse-Office	5/12/2022	12:30PM-3:30PM	0.20	4.00	0.10	20.30	79.40	
Scalehouse-Office	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.20	20.70	79.10	
Scalehouse-Office	10/12/2022	9:45AM-12:30PM	0.10	2.00	0.20	21.10	78.60	
Scalehouse-Office	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.20	21.00	78.80	
Scalehouse-Office	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.10	20.20	79.90	
Scalehouse-Office	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.10	20.00	79.90	
Maintenance Building	1/24/2018	9:30AM-12:30PM	0.10	2.00	0.20	22.50	77.10	doors closed fans on
Maintenance Building	5/21/2018	8:00AM-10:30AM	0.00	0.00	0.30	20.80	79.10	doors open
Maintenance Building	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.30	78.60	8/22 & 8/23 on field form, doors open and fans on
Maintenance Building	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.10	22.50	77.30	doors open, fans on
Maintenance Building	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.10	20.70	78.10	doors closed, fans on
Maintenance Building	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.10	20.80	79.00	doors open, fans on
Maintenance Building	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.10	21.30	78.60	doors open, fans on
Maintenance Building	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.20	22.80	76.80	doors closed, heater off, fan on
Maintenance Building	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.10	22.90	77.10	doors closed, heater off, fan on
Maintenance Building	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.10	21.60	78.30	doors closed, heater on, fan on
Maintenance Building	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.00	20.30	79.60	doors closed, fan on
Maintenance Building	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.10	22.20	77.70	doors closed, fan on
Maintenance Building	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.10	23.30	76.60	doors closed, fan on
Maintenance Building	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.10	21.50	78.40	doors closed, fan on
Maintenance Building	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.10	20.60	79.40	doors closed, fan on
Maintenance Building	10/29/2021	10:30AM-2:30PM	0.10	2.00	0.10	20.30	79.50	doors closed, fan on
Maintenance Building	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.30	22.90	76.80	doors closed, fan on
Maintenance Building	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.00	20.30	79.70	doors closed, fan on
Maintenance Building	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.10	20.80	79.10	doors closed, fan on
Maintenance Building	10/12/2022	9:45AM-12:30PM	0.10	2.00	0.10	21.30	78.40	doors closed, fan on
Maintenance Building	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.20	21.60	78.20	doors closed, fan on
Maintenance Building	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.10	20.00	79.80	doors closed, fan on
Maintenance Building	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.10	20.10	78.90	doors closed, fan on

1 - Values were recorded on the quarterly field forms.
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HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA

Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
Pump & Treat System Building	1/25/2018	9:30AM-12:30PM	0.00	0.00	0.10	21.90	76.80	
Pump & Treat System Building	5/21/2018	8:00AM-10:30AM	0.10	2.00	0.10	19.90	79.10	
Pump & Treat System Building	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.20	78.70	8/22 & 8/23 on field form
Pump & Treat System Building	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.10	21.20	78.70	ff, recorded values appear to be from previous
Pump & Treat System Building	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.10	21.20	78.70	ff, recorded values appear to be from previous
Pump & Treat System Building	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.10	21.20	78.70	ff, recorded values appear to be from previous
Pump & Treat System Building	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.10	21.20	78.70	door closed, recorded values appear to be from previous
Pump & Treat System Building	12/10/2019	9:00AM-12:30PM	NA	NA	NA	NA	NA	door closed
Pump & Treat System Building	3/17/2020	10:45AM-2:00PM	NA	NA	NA	NA	NA	door closed
Pump & Treat System Building	6/12/2020	10:45AM-2:00PM	NA	NA	NA	NA	NA	door closed
Pump & Treat System Building	7/16/2020	9:20AM-12:20PM	NA	NA	NA	NA	NA	door closed
Pump & Treat System Building	10/20/2020	10:00AM-1:00PM	NA	NA	NA	NA	NA	door closed
Pump & Treat System Building	1/22/2021	8:30AM-1:15PM	NA	NA	NA	NA	NA	door closed
Pump & Treat System Building	4/21/2021	12:00PM-2:30PM	NA	NA	NA	NA	NA	door closed
Pump & Treat System Building	7/21/2021	12:00PM-3:20PM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	10/29/2021	10:30AM-2:30PM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	3/10/2022	1:00PM-4:00PM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	5/12/2022	12:30PM-3:30PM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	8/25/2022	9:20AM-12:20PM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	10/12/2022	9:45AM-12:30PM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	2/8/2023	8:00AM-11:00AM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	4/13/2023	12:00PM-2:00PM	NA	NA	NA	NA	NA	building removed
Pump & Treat System Building	8/1/2023	7:00AM-9:30AM	NA	NA	NA	NA	NA	building removed
Leachate Management Building	1/26/2018	9:30AM-12:30PM	0.00	0.00	0.00	20.90	79.80	door closed
Leachate Management Building	5/21/2018	8:00AM-10:30AM	0.00	0.00	0.80	21.00	78.80	doors closed fans on
Leachate Management Building	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.20	79.80	8/22 & 8/23 on field form, doors open
Leachate Management Building	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.10	22.70	77.10	doors open
Leachate Management Building	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.20	21.40	78.30	doors closed, fans off
Leachate Management Building	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.10	20.80	79.00	door open, fan off
Leachate Management Building	9/10/2019	9:00AM-12:30PM	0.00	0.00	21.50	21.50	78.40	door open, fan off, CO2 appears anomalous
Leachate Management Building	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.20	19.40	80.20	door closed, no fan
Leachate Management Building	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.10	22.90	77.10	door closed, no fan
Leachate Management Building	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.10	21.70	78.30	door closed, no fan
Leachate Management Building	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.00	20.40	79.50	door open
Leachate Management Building	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.10	22.10	77.80	door closed
Leachate Management Building	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.10	22.60	77.30	door closed
Leachate Management Building	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.10	21.50	78.30	door closed
Leachate Management Building	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.10	21.50	78.30	door closed
Leachate Management Building	10/29/2021	10:30AM-2:30PM	0.00	0.00	0.10	21.00	78.90	door closed
Leachate Management Building	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.30	22.90	76.80	door closed
Leachate Management Building	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.00	21.20	78.80	door open
Leachate Management Building	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.10	20.80	79.10	door open
Leachate Management Building	10/12/2022	9:45AM-12:30PM	0.10	2.00	0.10	21.30	78.50	door open
Leachate Management Building	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.10	22.10	77.80	door open
Leachate Management Building	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.00	20.60	79.30	door open
Leachate Management Building	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.10	20.80	79.10	door open

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HISTORICAL PERIMETER LANDFILL GAS MONITORING DATA
Gas Migration Monitoring Plan
Wexford County Landfill

Location	Date	Time	Percent Methane (CH ₄) by Volume ¹	Methane Percent of LEL ²	Percent Carbon Dioxide (CO ₂) by Volume ¹	Percent Oxygen (O ₂) by Volume ¹	Balance ¹	Comments
Downwind	1/28/2018	9:30AM-12:30PM	0.10	2.00	0.10	21.20	77.00	
Downwind	5/21/2018	8:00AM-10:30AM	0.10	2.00	0.20	20.60	79.00	near front gate
Downwind	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.50	78.30	8/22 & 8/23 on field form
Downwind	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.10	22.10	77.70	near GP-8
Downwind	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.10	22.90	76.90	near GP-5
Downwind	6/14/2019	9:00AM-12:30PM	0.00	0.00	0.10	19.90	79.80	near front gate
Downwind	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.00	21.50	78.40	near front gate
Downwind	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.20	20.30	79.40	near GP-3R
Downwind	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.20	20.30	79.40	near GP-3R
Downwind	6/12/2020	10:45AM-2:00PM	0.10	2.00	0.10	23.00	76.80	near GP-3R
Downwind	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.10	20.60	79.30	near GP-2
Downwind	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.10	21.60	78.20	near GP-4
Downwind	1/22/2021	8:30AM-1:15PM	0.00	0.00	0.10	21.60	78.20	near GP-4
Downwind	4/21/2021	12:00PM-2:30PM	0.10	2.00	0.10	21.40	78.40	near GP-3R
Downwind	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.00	20.50	79.50	near GP-3R
Downwind	10/29/2021	10:30AM-2:30PM	0.00	0.00	0.10	20.60	79.30	near GP-3R
Downwind	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.10	22.40	77.60	near GP-5R
Downwind	5/12/2022	12:30PM-3:30PM	0.10	2.00	0.00	20.40	79.40	near front gate
Downwind	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.10	20.90	79.00	near GP-2
Downwind	10/12/2022	9:45AM-12:30PM	0.10	2.00	0.20	21.20	78.60	near front gate
Downwind	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.10	21.30	78.60	near GP-7
Downwind	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.10	19.80	80.10	near GP-7
Downwind	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.10	20.90	79.00	near GP-2
Upwind	1/27/2018	9:30AM-12:30PM	0.10	2.00	0.00	21.50	78.30	
Upwind	5/21/2018	8:00AM-10:30AM	0.10	2.00	0.10	20.60	79.10	near GP-2
Upwind	8/22/2018	7:30AM-10:30AM	0.00	0.00	0.10	21.20	78.70	8/22 & 8/23 on field form
Upwind	10/22/2018	10:00AM-4:40PM	0.00	0.00	0.10	22.20	77.60	near GP-3R
Upwind	2/21/2019	9:00AM-11:30AM	0.00	0.00	0.20	20.30	79.50	near GP-2
Upwind	6/14/2019	9:00AM-12:30PM	0.10	2.00	0.10	20.60	79.20	near GP-2
Upwind	9/10/2019	9:00AM-12:30PM	0.00	0.00	0.30	18.00	79.00	near GP-2
Upwind	12/10/2019	9:00AM-12:30PM	0.00	0.00	0.10	22.00	77.70	near front gate
Upwind	3/17/2020	10:45AM-2:00PM	0.00	0.00	0.10	22.00	77.70	near front gate
Upwind	6/12/2020	10:45AM-2:00PM	0.00	0.00	0.00	20.90	79.10	near GP-8
Upwind	7/16/2020	9:20AM-12:20PM	0.00	0.00	0.10	20.70	79.30	near front gate
Upwind	10/20/2020	10:00AM-1:00PM	0.00	0.00	0.10	22.10	77.80	near BH-1
Upwind	1/22/2021	8:30AM-1:15PM	0.00	0.00	2.40	21.10	76.50	near BH-1
Upwind	4/21/2021	12:00PM-2:30PM	0.20	4.00	0.10	20.80	78.90	near front gate
Upwind	7/21/2021	12:00PM-3:20PM	0.00	0.00	0.10	20.10	79.80	near front gate
Upwind	10/29/2021	10:30AM-2:30PM	0.10	2.00	0.10	20.10	79.70	near front gate
Upwind	3/10/2022	1:00PM-4:00PM	0.00	0.00	0.10	21.70	78.20	near front gate
Upwind	5/12/2022	12:30PM-3:30PM	0.00	0.00	0.00	20.20	79.80	near GP-2
Upwind	8/25/2022	9:20AM-12:20PM	0.00	0.00	0.10	20.70	79.20	near front gate
Upwind	10/12/2022	9:45AM-12:30PM	0.10	2.00	6.00	15.10	78.80	near GP-2
Upwind	2/8/2023	8:00AM-11:00AM	0.00	0.00	0.10	21.60	78.30	near BH-2
Upwind	4/13/2023	12:00PM-2:00PM	0.00	0.00	0.00	20.20	79.80	near GP-1
Upwind	8/1/2023	7:00AM-9:30AM	0.00	0.00	0.10	20.30	79.60	near GP-7

1 - Values were recorded on the quarterly field forms.
2 - Values were calculated based on values recorded on the quarterly field form.

Gas Migration Monitoring Plan Review Checklist			
Site Name: <u>Wexford County Landfill</u>		Review Date: <u>October 16, 2023</u>	
WDS Number: <u>470336</u>		Date of Plan: <u>October 2023</u>	
Comments on sufficiency of coverage ↓↓↓↓↓↓			
Is there sufficient coverage of probes to protect nearby offsite structures?	YES		
Does the site include continuous monitoring in each onsite structure?	YES	Does the site provide any offsite continuous monitors?	NO
What kind of monitor is utilized?	MSA Ultima Gas Detector	What kind of monitor is utilized?	NA
How often are meters tested/calibrated?	Quarterly	How often are meters tested/calibrated?	OTHER
How frequently are they replaced?	As Needed	How frequently are they replaced?	NA
How many gas monitoring probes are there?	7		
Is there an engineering detail of the probes?	YES	Do the installed probes comply with the detail?	YES
Do all probes extend to the uppermost water table or at least 10' below waste grade?	YES	GP-8 is screened a minimum of 10 feet below waste grade for currently constructed cells, however, based on future construction, GP-8 may not be screened a minimum of 10 feet below grade	
If not please explain each deviation:	=====>		
Are probes fully screened in vadose zone? (not small discreet intervals)		YES	
If not please explain each deviation:	=====>		
What is the size of each probe boring?	minimum 6"	What is the size of the probe piping?	2" diameter
What is the annular space / well pack filled with?	Number 30 coarse sand or 10/20 sand pack		
Are gas probes purged prior to sampling?	YES	Volume purged:	250 mL
What meter is utilized for probe measurement?	GEM 2000 in the past year	Are meters calibrated or bump tested prior to quarterly testing?	YES
Is reading taken in accordance with meter manufacturer's recommendations (equipment and hosing purge)	YES		
Does the Gas Migration Monitoring Plan include a monitoring checklist/form to be filled out each monitoring event?	YES		
Does the monitoring checklist/form include Date, Time, Location, Purge time/volume, CH ₄ level, LEL %, O ₂ at each location	YES		
Are there any previous migration issues at the landfill?	YES		
If so, how have they been addressed and how are they monitored (i.e., out of waste collectors, more frequent monitoring, etc.)?	On October 29, 2021, an exceedance of the LEL was noted at GP-4R. Three barnholes were completed to the east of GP-4R (towards the property boundary) were punched and monitored on October 29, 2021 and no methane was detected. On October 12, 2022, two perimeter gas probes (GP-4R and GP-5R) had exceedances noted based on a review of the quarterly perimeter landfill gas monitoring data for the past five years. At the time of the perimeter landfill gas monitoring, the gas extraction and flare system was turned off for repairs. The system was back on the following week, and verification monitoring was conducted on October 19 and October 20, 2022 and the field staff recalls no exceedances. Unfortunately, the documentation of the field readings from the October 19 and 20, 2022 verification event has not been located, but subsequent monitoring events have reported no detections of methane in GP-4R or GP-5R, indicating that the detections reported on October 12, 2022 were directly related to the inactivity of the gas extraction and flare system.		
Does the current plan address previous migration issues and do the elements meet the requirements of 324.11512b through 11512h?	Since the gas migration exceedance was not confirmed by the boreholes or subsequent monitoring events, this plan does not discuss previous migration issues. This plan describes additional monitoring steps if migration issues are observed. Elements related to potential additional gas extraction requirements will be included in the Gas Collection and Control Plan (GCCP).		